

Generative AI Course Syllabus

Archer Infotech, Kothrud Pune | Classroom & live-online batches | Updated September 2026

Course Overview

This Generative AI syllabus is designed for students, developers, working professionals, and business users who want to understand and apply modern AI tools in practical workflows. The course starts with the foundations of artificial intelligence and moves toward prompt engineering, large language models, image generation, embeddings, RAG, AI agents, automation, responsible AI, and project-based implementation.

Prerequisites

For Non-Coding Learners

- Basic computer and internet usage
- Ability to use browser-based tools, email, documents, and spreadsheets
- Basic English reading and writing for prompt creation
- Curiosity to explore AI tools through hands-on practice
- Basic understanding of business, education, content, or office workflows

For Technical Learners

- Basic programming knowledge in Python, JavaScript, or Java
- Basic understanding of HTTP, REST APIs, and JSON
- Git and GitHub basics
- Basic database awareness: SQL and simple NoSQL concepts
- Basic web development understanding
- Optional but useful: command line, Docker, cloud basics, Spring Boot, Node.js, or Python backend development

Syllabus Index

Part 1: Foundations of Generative AI

- Introduction to Artificial Intelligence
- AI vs Machine Learning vs Deep Learning vs Generative AI
- Evolution of Generative AI
- Real-world Use Cases of Generative AI
- Popular Generative AI Tools
- Opportunities and Limitations of Generative AI

Part 2: Large Language Models

- What is a Large Language Model
- How LLMs Understand and Generate Text
- Tokens, Context Window, Temperature, and Parameters
- Model Training, Fine-tuning, and Inference
- Common LLM Capabilities
- Common LLM Limitations and Hallucinations

Part 3: Prompt Engineering

- Introduction to Prompt Engineering
- Writing Clear and Effective Prompts
- Role-based Prompting
- Zero-shot, One-shot, and Few-shot Prompting
- Prompt Templates and Reusable Prompt Patterns
- Prompting for Coding, Debugging, Writing, Research, and Analysis

- Prompt Evaluation and Iteration

Part 4: Generative AI for Text and Content Creation

- Content Ideation and Drafting
- Summarization and Rewriting
- Email, Blog, Social Media, and Marketing Content
- Documentation and Technical Writing
- Translation and Localization Support
- AI-assisted Study Notes and Learning Material

Part 5: Generative AI for Images, Audio, and Video

- Introduction to Multimodal AI
- Text-to-Image Generation
- Image Editing and Style Transfer
- AI-assisted Presentation and Design Workflows
- Audio Generation and Transcription Basics
- Video Generation and Editing Concepts

Part 6: Generative AI for Developers

- AI-assisted Coding Workflows
- Code Generation and Refactoring
- Debugging with AI
- Test Case Generation
- API Documentation Generation
- Reviewing AI-generated Code

Part 7: Embeddings and Vector Search

- What are Embeddings
- Semantic Search vs Keyword Search
- Vector Databases and Vector Stores
- Similarity Search Concepts
- Chunking and Document Processing
- Knowledge-base Use Cases

Part 8: Retrieval-Augmented Generation

- Introduction to RAG
- Why RAG is Needed
- RAG Pipeline Architecture
- Document Loading and Chunking
- Embedding Documents
- Retrieving Relevant Context
- Generating Answers with Sources
- Evaluating RAG Output

Part 9: AI Agents and Automation

- What are AI Agents
- Agent vs Chatbot
- Tools, Memory, Planning, and Actions
- Function Calling Concepts
- Workflow Automation with AI
- Risks and Controls in Agentic Systems

Part 10: Generative AI APIs and Application Integration

- Introduction to AI APIs
- Chat Completion and Response APIs

- Prompt Templates in Applications
- Structured Output Concepts
- Building AI Features in Web Applications
- Logging, Cost Control, and Rate Limits

Part 11: Responsible AI and Security

- Bias, Fairness, and Ethical AI
- Hallucination and Output Verification
- Privacy and Data Protection
- Prompt Injection Basics
- Secure AI Application Design
- Human Review and Approval Workflows

Part 12: Capstone Projects

- AI Resume Analyzer
- AI Interview Preparation Assistant
- AI Document Question-Answering System
- AI Customer Support Chatbot
- AI Content Generator for Social Media
- AI Knowledge Base with RAG

Recommended Additions for Industry Readiness

These topics strengthen the syllabus for real-world AI application development and should be included as practical sessions, workshops, or project checkpoints.

AI Product Thinking

- When to use AI and when not to use AI
- AI feature design and user experience
- Business value, cost, latency, and reliability trade-offs
- Designing AI workflows with human review

Model Selection and Cost Management

- Choosing models for speed, quality, reasoning, multimodal use, and cost
- Token estimation and context-window planning
- Rate limits, retries, caching, and fallback models
- Cost tracking for AI-powered applications

Structured Outputs and Function Calling

- JSON output design
- Schema validation
- Function calling concepts
- Tool argument validation
- Reliable extraction from unstructured text

Evaluation and Quality Improvement

- Prompt evaluation
- RAG evaluation
- Hallucination checks
- Human review rubrics
- Regression testing for AI workflows
- Improving prompts and pipelines using test cases

AI Security and Guardrails

- Prompt injection awareness

- Sensitive data leakage prevention
- Output validation and moderation
- PII filtering
- Tool-use boundaries
- Audit logs and safe logging practices

Deployment and Monitoring

- Logging prompts and responses safely
- Monitoring errors, cost, latency, and user feedback
- Versioning prompts and AI workflows
- Production readiness checklist for AI applications

Learning Outcomes

- Explain generative AI, LLMs, prompts, embeddings, RAG, and agents
- Use AI tools effectively for learning, productivity, content, and development
- Write strong prompts for real-world tasks
- Evaluate and improve AI-generated output
- Build simple AI-assisted applications and automation workflows
- Understand responsible AI, privacy, hallucination, and prompt-injection risks

What to Learn After Course Completion

AI Application Developer Path

- Python FastAPI or Node.js backend development
- OpenAI, Anthropic, Gemini, or similar AI APIs
- Vector databases and production RAG applications
- AI SaaS project development

Java and Spring AI Path

- Spring Boot REST APIs
- Spring AI
- PostgreSQL with pgvector
- Java-based RAG applications
- Enterprise AI integration patterns

Agentic AI Developer Path

- LangGraph, LangChain, CrewAI, or OpenAI Agents SDK
- Tool calling and workflow orchestration
- Tracing and evaluation
- Multi-agent workflows

AI Automation Path

- Zapier, Make, n8n, or similar workflow tools
- AI-powered CRM, email, HR, education, and support automation
- No-code and low-code AI workflow design

AI Security and Governance Path

- OWASP Top 10 for LLM and Generative AI Applications
- NIST AI Risk Management Framework
- Google Secure AI Framework
- Prompt-injection testing and AI risk assessment

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